

Towards Quantitative Assessment of Performance Degradation in Model-Based Galileo SAR Systems Considering Hypothermia and Physiological-Psychological Variability

Abstract

The Galileo Search and Rescue (SAR) system is a critical European maritime rescue operation that uses a satellite network to locate individuals in distress. Designed for longevity (over 12 years), its stable components adhere to fundamental trustworthiness and dependability parameters such as Reliability, Availability, and Maintainability (RAM), which are essential for evaluating rescue performance. This paper presents a novel approach that combines Continuous-Time Markov Chains (CTMCs) and Concurrent Stochastic Games (CSGs) in a two-phase approach where operational and performance properties are expressed using both formalisms. Continuous Stochastic Logic (CSL) and reward Probabilistic Alternating Temporal Logic (rPATL) are employed to evaluate the performance of the SAR communication service over the operational time, while considering human hypothermia and physiological-psychological degradation. We rely on Words of Estimative Probability (WEPs) to map the level of human status degradation, based on clinical resources found in the literature. We leverage the classical PRISM model checker and PRISM-games for quantitative analysis, focusing on degradation scenarios within the SAR components and the evolving status of the distressed person. This study identifies the primary degradation scenarios that impact the quality of the rescue mission, considering the human status, and then demonstrates how human survival attempts can be modeled within such scenarios.

Keywords: Galileo Search and Rescue, Reliability, Performance, Stochastic Games, PRISM

Contents

1	Introduction	3
2	Related work	5
2.1	Human behavior in Cyber-physical systems	5
2.2	Satellite systems	5
3	Theoretical Background	6
3.1	The PRISM language for Markov models	6
3.1.1	Continuous-Time Markov Chains (CTMCs)	6
3.1.2	Concurrent Stochastic Games (CSGs)	6
3.2	Quantifying human decisions	7
4	Physiological and Psychological Effects of Hypothermia	7
4.1	The hypothermia stages	7
4.2	A formal model of the hypothermia stages	8
5	Communication System Modeling in Galileo Satellites	10
5.1	The SAR system model	12
5.2	Operational capabilities characteristics	13
5.3	A formal model of satellite communication degradation	15

23	6 Quantitative Analysis using PRISM	16
24	6.1 Galileo satellite system dependability	16
25	6.1.1 Safety	16
26	6.1.2 Reliability	18
27	6.1.3 Operational rescue availability	19
28	6.2 Human-in-the-Loop satellite system dependability	20
29	6.2.1 Reliability	21
30	6.2.2 Operational rescue availability	23
31	6.3 Discussion	23
32	6.3.1 Lessons Learned from Human-in-the-Loop	23
33	6.3.2 From Markov model to game-theoretical model	24
34	6.4 Threats to validity	24
35	7 Conclusion	26

1. Introduction

Satellite systems are essential in various sectors, including maritime [1] and civil operations [2]. The increasing global demand for reliable communication has driven significant innovation within the space industry [3]. Given the critical nature of these systems, dependability assurance is paramount, including the performance assessment of operational tasks, particularly those related to human rescues, such as search and rescue (SAR) missions [4, 5, 6].

Formal verification [7, 8] is a powerful technique that allows for thorough system analysis to formally prove a wide range of properties, ensuring its correct operation, even in the presence of stochastic phenomena. Various formalisms can be employed, each tailored to specific applications. Common examples include Markov Decision Processes (MDPs) [9], Continuous-Time Markov Chains (CTMCs) [10], and Concurrent Stochastic Games (CSGs) [11]. In contrast to simulation, which relies on analyzing results from many random samples, formal verification provides a mathematically rigorous and exhaustive analysis of the system’s behavior.

Our work illustrates how to effectively model SAR communication efficiency using Galileo satellites and verify its performance under degradation effects using the PRISM model checker [8]. Although PRISM has been widely applied to verify the correctness and effectiveness of hardware and software designs [12], its application to the specific context of SAR systems has not been addressed. Previous research [13, 14, 15, 16] has primarily focused on assessing the dependability of the satellite itself (e.g., the US GPS Satellite). In contrast, this work centers on the performance of communication services during request-response interactions between a person in distress and the components of the Galileo satellite system. This system handles requests, communicates with ground services for processing, and dispatches rescue personnel to assist individuals in distress. We consider various degradation sources as reported in the official documentation [4, 5, 6]. Specific anomalies, such as loss of communication with the ground station, can result in severe degradation of SAR services. These anomalies are identified through availability monitoring, as described in the same documentation. Formally, the SAR system model is specified as a Continuous-Time Markov Chain (CTMC) [17]. Assuming constant failure and repair rates for the SAR communication services, both the time to failure and the time to repair are modeled as exponentially distributed random variables.

The characteristics of a Galileo SAR system do not solely determine the reliability of a rescue operation; the condition of the person being rescued also plays a critical role. Human degradation factors, encompassing both physiological elements (like fatigue, hypothermia or hyperthermia, dehydration, and injury) and psychological elements (such as stress, panic, anxiety, fear, cognitive load, and despair), can significantly impair the effectiveness of rescue signal transmission and the overall response [18]. Unlike earlier models primarily focused on technological aspects, this research incorporates human behavior as a *sensing element* within the system [19]. In our study, we assume that physiological stressors, such as hypothermia, can impair a person’s cognitive load, affecting the rate at which they activate the distress signal. This assumption is based on clinical work [20, 21]. The severity of hypothermia is expressed using the Kent classification (Words of Estimative Probability) [22], a method also employed in studies for quantifying risks or observations stemming from expert judgment. This quantitative property enables the effective integration of observations into CTMC models, particularly in identifying hypothermia stages.

We employ a complementary approach that uses Concurrent Stochastic Games (CSGs) to extend beyond the traditional Markov model in CTMC [23]. This offers a new perspective, transitioning from classical Markov models to a game-theoretic model. This model examines the dynamic interplay between the efforts of SAR operators to ensure the reliability of rescue for individuals seeking assistance. The person to be rescued will attempt to survive through multiple efforts (e.g., muscle movement), unaware of the satellite system’s behavior, while the system simultaneously tries to maintain the efficiency of the rescue mission. This perspective differs from classical Markov models, which typically express collaborative behavior.

Approach. The summary of our approach is portrayed in Figure 1. Initially, we integrate human parameters related to their physiological and psychological status to construct a stage-based model. The human’s exposure to hypothermia determines this variability. The hypothermia parameters, sourced from clinical

literature, are then mapped to the Kent classification (Words of Estimative Probability) [22] as probabilities ①, representing the person's reliability in response to a critical situation (e.g., muscle movement). Subsequently, the different stages of hypothermia are linked through temporal properties of those stages ②. These stages lead to a state-based model that can be interpreted in CTMC ③. In the second phase, the interactions between the Galileo SAR System components are modeled using Continuous-Time Markov Chains (CTMCs), enriching the model with reliability and availability parameters (dependability) ④. The CTMC model is first constructed independently of human body hypothermia. Then, it is composed with the human stage-based model ⑤ and subsequently verified against Continuous Stochastic Logic (CSL) properties, which represent dependability properties ⑥. The collected analysis results lead to a new modeling perspective where Concurrent Stochastic Games (CSGs) are employed, with the identification of two players ⑦: the person in distress and the satellite system. This model incorporates survival attempts to enhance the rescue mission, as humans often prioritize self-rescue. The model is then verified ⑧ against dependability properties expressed in probabilistic alternating-time temporal logic with rewards (rPATL) using the PRISM-games model checker ⑨. The comparison between the CTMC and CSG models is based on the delta (difference) generated from the results of both models for reliable mission rescue accomplishment.

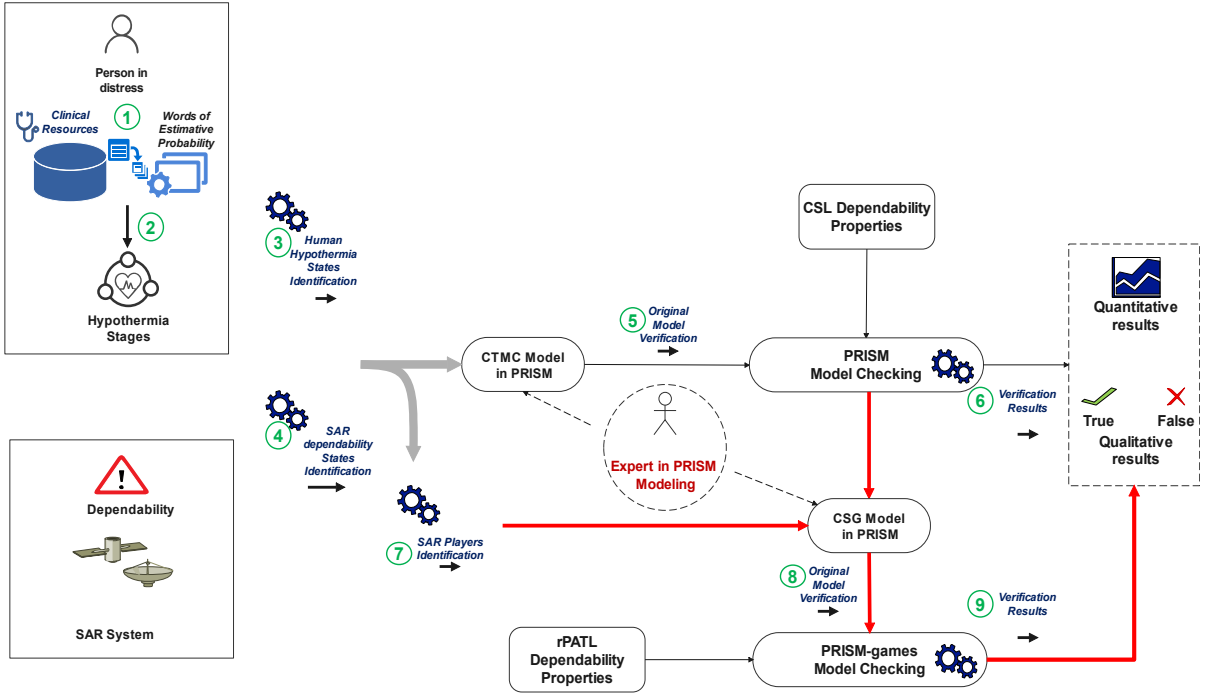


Figure 1: Overview of the Approach for Model-Based Galileo SAR Systems Considering Human Factors.

Outline. The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 provides a brief theoretical background on CTMCs and CSGs in PRISM. Section 4 provides the construction steps for hypothermia stages, starting from physiological and psychological variability. Section 5 presents our proposed modeling approach for SAR systems, identifying all communicating components. We then perform a quantitative analysis of the SAR service scenarios in Section 6. Finally, Section 7 concludes the paper and suggests directions for future research.

2. Related work

In this section, we will address related research on human behavior in cyber-physical systems and the role of verification in satellite systems. This aims to provide readers with sufficient context regarding the integration of these two crucial domains.

2.1. Human behavior in Cyber-physical systems

The relationship between humans and cyber-physical systems has been extensively documented in the literature [24, 25] since the Second World War. Early foundations can be traced to Norbert Wiener’s “Cybernetics: Or Control and Communication in the Animal and the Machine” in 1948 [26], where a good example is mentioned in the book about an aviator under combat stress is rarely capable of complex, unconstrained voluntary behavior, and will likely default to pre-trained activity patterns dictated by their interaction with the aircraft. The book by Kirwan [27] developed a groundbreaking methodology to assess human reliability during interaction with Cyber-Physical Systems (CPS). In this framework, Human Reliability Assessment (HRA) is primarily derived from human error factors and their quantification. The methodology mainly focuses on errors stemming from human behavior and its application in power plants, nuclear facilities [28], or crane orchestration [29]. However, what if the sources of errors are internal to the human, not directly controlled, but influenced by environmental factors like heat and cold? Such factors could impact human behavior, potentially leading to actions like pushing the wrong button or definitively failing, as addressed in the paper.

2.2. Satellite systems

Several works in the literature have addressed the Performance and reliability of satellite systems. In [15], Peng et al. propose using CTMC to model satellite systems, demonstrating how this approach can analyze the impact of various factors, such as solar radiation, on satellite reliability and maintenance. Building upon this work, Hoque et al. [13] incorporates Erlang distributions into the CTMC framework to further refine the modeling of maintenance and scrubbing activities. Additionally, Baouya et al. [16] broadens the scope by incorporating human capabilities in maintaining satellite systems within a game-theoretical model.

While significant research has focused on the quality and deployment of US GPS satellites, research activities explicitly addressing the reliability and performance of the SAR/Galileo satellite system are less prevalent in the literature. Studies addressing this topic include [30, 31, 32]. In [32], Lewandowski et al. investigate a hybrid communication infrastructure combining terrestrial networks with enhanced Galileo SAR for firefighter missions. Using simulations, they compare the advanced Galileo SAR system with the existing Cospas-Sarsat system. They identify performance metrics and determine the optimal number of Galileo satellites to enhance SAR service quality and minimize emergency response times. Andrei et al. [33] evaluate the performance of four Galileo satellites launched in 2018 during their first year of operational service (to reach 22 satellites). Key performance indicators were analyzed, including signal-in-space status, availability, and ranging accuracy. Results demonstrate high signal health and availability, with minimal signal interruptions. Moreover, the numerical results demonstrate high reliability for the most recent additions to the Galileo satellite constellation. The authors in [31] focus on developing the SAR/Galileo ground system in Korea, outlining a roadmap strategy for the next-generation search and rescue system to enhance rescue efficiency by improving location accuracy and reducing response times. The authors in [30] investigate methods to improve the Return Link Service (RLS) availability in the Galileo SAR system. Their proposed solutions utilize Network Coding to improve RLM reception by mitigating signal losses due to harsh weather or obstacles, considering backward compatibility and system complexity.

Most existing approaches primarily focus on the technological aspects of satellite system components and their mission performance, often neglecting human operators’ physiological and psychological variability. Incorporating these human factors can provide designers with insights into mission reliability. For instance, the foundational work by Swain and Guttman [18] highlights many key factors that impair physical infrastructure from performing reliably. Specifically, human vigilance can be significantly affected if an operator experiences insufficient arousal or is prone to fatigue.

This paper offers a mathematical and rigorous approach to verify the influence of human behavior and status (stemming from physiological and psychological states) in the context of external environments like cold. This influence is then composed to cyber-physical systems (including satellite systems and signal quality). The study addresses such behavior from two perspectives: analyzing the evolution of behavior over time using CTMCs, and from a collaborative standpoint where humans interact with physical systems, driven by the instinct of survival and seeking rescue in crises involving cold water in CSGs.

3. Theoretical Background

In this section, we will revisit the theoretical foundations of Markov models, such as CTMCs, and game models, specifically CSGs, interpreted in the PRISM language [8]. Additionally, we will revisit the concept of Words of Estimative Probability (WEPs) to demonstrate how decisions are represented in a probabilistic manner.

3.1. The PRISM language for Markov models

Probabilistic Model Checking using PRISM [8] relies on constructing a formal model represented using appropriate storage structures. The verification process is then performed by applying a suite of algorithms implemented within the PRISM engine [34]. For our analysis, we employ Continuous-Time Markov Chains (CTMCs) [17], a well-established modeling technique for evaluating reliability and performance, and Concurrent Stochastic Games (CSGs) [23] to model the stochastic interplay between communication entities.

The PRISM model is composed of a set of modules that can synchronize. Each module is characterized by variables and commands (or transitions). The valuations of these variables represent the state of the module. The behavior of each module is described using a set of commands, each of which follows the following format:

$$[a] \ g \rightarrow \lambda : u$$

This indicates that if the guard condition g evaluates to true, then the update u is enabled to occur with a rate of λ for action a . A guard is a boolean formula constructed from the module variables. The update u is an evaluation of variables expressed as a conjunction of assignments: $v'_i = val_i + \dots + v'_n = val_n$ where $v_i \in V$, with V being a set of local and global variables, and val_i are values evaluated via expressions denoted by θ such that $\theta : V \rightarrow \mathbb{D}$, where $\mathbb{D}$ is the domain of the variables.

3.1.1. Continuous-Time Markov Chains (CTMCs)

The CTMC involves a set of states and a transition matrix $\mathbf{R} : S \times S' \rightarrow \mathbb{R}_{\geq 0}$. The rate specifies the delay before a transition between states s and s' takes place $\mathbf{R}(s, s')$, where the probability between s and s' take within time t is given by the value $1 - e^{-\mathbf{R}(s, s') \times t}$. Based on research using PRISM for CTMC modeling, as outlined in [35], exponentially distributed delays are often considered suitable for modeling electronic component lifetimes and inter-arrival times.

Properties are expressed in Continuous Stochastic Logic (CSL) [36]. For instance, the following property described in natural language: *Is the probability of that eventually the system failure occurring within 100 time units is less than 0.001* is expressed as: $P_{<0.001}[\mathbf{F}^{\leq 100} \text{fail}]$ Here, *fail* is the label that refers to the system failure states. Regarding the reward structure, the property expressed in natural language: *What is the amount of reward accumulated over a specific 100 times units ?* is expressed in CSL as: $\mathbf{R}\{\text{"up"}\} = ?[\mathbf{C} \leq 100]$.

3.1.2. Concurrent Stochastic Games (CSGs)

CSG [23] is a PRISM model with formalism based on the idea that players make choices concurrently in each state and then transition simultaneously. Each player controls one or more modules, and the actions that label commands within a player's modules must only be used by that specific player.

The properties related to CSGs are expressed in the temporal logic rPATL [37] (short for reward Probabilistic Alternating Temporal Logic). The property grammar is based on CTL [7], extended with the

coalition operator $\langle\langle C \rangle\rangle$ of ATL [38] and the probabilistic operator P of PCTL [39]. For instance, the following property expressed in natural language: *Players 1 and 2 have a strategy that guarantees the probability of computer shutdown within 100 rounds remains below 0.001, even in the presence of memory failures* is expressed in rPATL as: $\langle\langle 1, 2 \rangle\rangle P_{<0.001} [F (\text{fail} \ \& \ \text{rounds} = 100)]$. Here, **fail** is the label that refers to the system failure states. Regarding the reward structure, the property expressed in natural language: *What is the reward r within 100 rounds to reach fail for both Players 1 and 2 for a selected strategy?* is expressed in rPATL as: $\langle\langle 1, 2 \rangle\rangle R = ? [F (\text{fail} \ \& \ \text{rounds} = 100)]$.

3.2. Quantifying human decisions

Words of Estimative Probability (WEPs) or Kent-like classification scheme, as outlined in [22], are terms intelligence analysts use in their reports to communicate the likelihood of a future event occurring. Their application extends to various fields, including medicine [40], policy analysis, intelligence, and within the reports of the Intergovernmental Panel on Climate Change [41]. According to [22], this scheme relies on the information provided in Table 1, where each level of certainty is associated with a likelihood (i.e, probability) to inform about human status or their level of decision.

Certainty	Percentage	Margin of Error
Certain	100%	0%
Almost Certain - “likely”	93%	6%
Probable	75%	12%
Chances About Even	50%	10%
Probably Not	30%	10%
Almost Certainly Not	7%	5%
Impossible	0%	0%

Table 1: Kent Probability Classification [22]

The application of Words of Estimative Probability (WEPs) in medicine often relies on a doctor’s experience and background. For instance, a doctor might state: *“Based on the patient’s symptoms and the lab results, it is probably that the patient has strep throat.”* Similarly, in climate science, when measuring planetary warming, analyses might conclude: *“Global mean surface temperature will likely rise by 1.5°C to 4.5°C by 2100 under current emissions pathways, leading to increases in extreme heat events.”* In the context of cyber-physical attacks, it could be assessed: *“Without adequate physical hardening and monitoring, an edge server deployed in a publicly accessible environment will probably be physically tampered with, leading to potential data compromise or service disruption.”*

4. Physiological and Psychological Effects of Hypothermia

Hypothermia is defined as a drop in body temperature below 35 °C. Hypothermia develops when the body loses heat faster than it can produce or retain it, leading to an inability to maintain homeostasis and normal physiological functions [21]. In this section, we review the different stages of hypothermia, including its physiological and psychological effects, as well as the estimated transition times between them. Additionally, we propose a formal model based on Markov processes that permits mathematical reasoning.

4.1. The hypothermia stages

Five stages have been defined in the paper. The first stage is characterized by a body temperature between 32 °C and 35 °C, during which clear consciousness with shivering is observed, accompanied by slowed breathing and heart rate, slurred speech, a decline in mental function, and hallucinations [42]. In Stage 2, when the body temperature is between 28 °C and 32 °C, impaired consciousness without shivering is observed. At this level, heart rates may be significantly below normal, and atrial ectopy and atrial fibrillation frequently appear [43]. Stage 3 is characterized by a temperature below 28 °C but at or above 24

°C, at which point the person is unconscious. At this stage, symptoms often include loss of shivering, low blood pressure (hypotension), fluid in the lungs, absence of reflexes, complete muscle stiffness, and loss of voluntary motion [42]. At Stage 4, the person is in a coma and has a Loss of voluntary motion. The latest stage shows death due to irreversible hypothermia when the temperature is less than 13.7 32 °C. The study will not concern this stage.

Table 2 presents a synthesis of works from [44, 45, 42] that allows us to infer the possible movements of individuals in critical situations exacerbated by hypothermia. We sought to approximate these probabilities based on relevant clinical results. For instance, in Stage 1 of hypothermia, it is very likely that the person still retains the capability for voluntary movement. However, in Stage 2, it is probable that the person still has the use of their arms. Given that a person in Stage 3 is unconscious, voluntary arm movement is practically absent. Furthermore, in Stage 4, where the person is in a coma, the use of their arms is almost certainly no longer possible.

According to a clinical study by [46], in very cold water (2°C), the core temperature is expected to drop at a rate of 0.35°C/min. However, research by [47] indicates that, from an initial core temperature of 37°C, it drops to 35°C within approximately 25 minutes. Based on these reported rates, we can estimate the time required to transition between different stages of hypothermia. Therefore, the estimated times for stage transitions are as follows:

- Mean time from Normal state (37°C) to Stage 1 (35°C), MTS_{h1} : 25.00 minutes.
- Mean time from Stage 1 (35°C) to Stage 2 (32°C), MTS_{h2} : 8.57 minutes
- Mean time from Stage 2 (32°C) to Stage 3 (28°C), MTS_{h3} : 11.43 minutes
- Mean time from Stage 3 (28°C) to Stage 4 (24°C), MTS_{h4} : 11.43 minutes

4.2. A formal model of the hypothermia stages

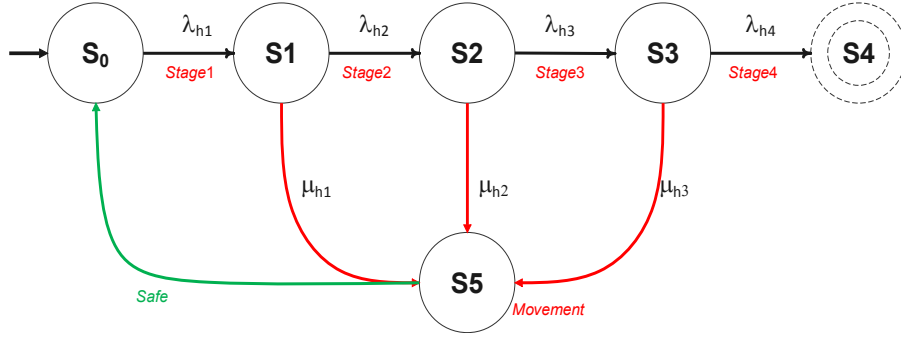


Figure 2: CTMC Model for Hypothermia Stages and Movement.

Building upon the physiological and psychological synthesis of hypothermia, we can compute the rate of human degradation. This approach is analogous to how degradation rates are modeled for physical systems, where each status of degradation is typically associated with a specific rate as follows:

$$\forall i \in \mathbb{N}, 1 \leq i < 5, \quad \lambda_{hi} = \frac{1}{MTS_{hi}} \quad (1)$$

Considering the equation in 1, the CTMC model in Figure 2 captures the different stages of hypothermia – S_1, S_2, S_3, S_4 – which are linked and labeled by λ_{hi} modeling the rate of stage occurrence. State S_0 models the normal status of a human before immersion in cold water.

Stage	Core Temp (°C)	Consciousness	Shivering	Voluntary movement (incl. arms)	Probability of voluntary arm use (WEP)
HT I	35–32	Conscious	Present	Yes, often intact	Very likely
HT II	<32–28	Impaired consciousness	±	Variable (frequent in-coordination)	Probable
HT III	<28–24	Unconscious	Absent	No	Almost Certainly Not
HT IV	<24	Coma / asystole	Absent	None	Nearly impossible ($\approx 0\%$)

Table 2: Estimated probability of voluntary arm use by hypothermia stage, based on clinical signs and mapped using Words of Estimative Probability (WEP).

As we consider that a human can use their arms, we can model such behavior as a possible transition from S_1, S_2, S_3 to state S_5 . This represents the ability to raise a hand or act on specific devices to alert for help, signifying the human's attempt to self-rescue. These transitions are labeled by μ_{hi} as the rate of such self-saving actions (analogous to a repair rate, like Mean Time To Repair, in cyber-physical systems). At the stage S_4 , the human is not able to raise a hand according to Table 2.

The probability of voluntary arm movement depends on the parameters presented in Table 2. This can be considered analogous to a system's availability (up/down states) in cyber-physical systems. The probability of such hand activation is calculated as:

$$\forall i \in \mathbb{N}, 1 \leq i < 4, \quad P = \frac{\mu_{hi}}{\lambda_{hi} + \mu_{hi}} \quad (2)$$

Based on the probability of voluntary arm movement, the formula for computing μ_i can be expressed as follows:

$$\forall i \in \mathbb{N}, 1 \leq i < 4, \quad \mu_{hi} = \frac{P \times \lambda_{hi}}{1 - P} \quad (3)$$

Finally, individuals suffering from hypothermia in Stages 1, 2, and 3 can be saved by a rescue operation. This type of rescue can be modeled as a transition from S_5 to S_0 , as molded in Figure 2.

The Listing 1 provides the PRISM code associated with the Markov model schematized in Figure 2. The PRISM code begins with constant definitions that represent the different state values, aiding in the identification of variables within the PRISM modules. These constants also include references to calculated rates, such as the rate of progress from different hypothermia stages (λ_{hi}) in lines 11-14, and recovery rates associated with human physical action capability (μ_{hi}) in lines 17-19. Parameters related to degradation rates and recovery are presented in Table 3.

Parameter Description	Mean Time (minutes)	Rate
Normal (37°C) to Stage I (35°C)	25.00 (fixed)	$\lambda_{h1} = 0.0400$
Stage I (35°C) to Stage II (32°C)	8.57	$\lambda_{h2} = 0.1167$
Stage II (32°C) to Stage III (28°C)	11.43	$\lambda_{h3} = 0.0875$
Stage III (28°C) to Stage IV (24°C)	11.43	$\lambda_{h4} = 0.0875$
Rescue from Stage I	–	$\mu_{h1} = 1.5503$
Rescue from Stage II	–	$\mu_{h2} = 0.2625$
Rescue from Stage III	–	$\mu_{h3} = 0.0066$

Table 3: Hypothermia Model Parameters

The transitions are identified in lines 21-69. Specifically, Line 30 identifies the transition from the normal state to the first stage of hypothermia. Line 37 identifies the transition from the first stage of hypothermia to the second stage. Line 46 models a human performing physical actions (self-rescue). Line 49 permits transition from the second stage of hypothermia to the third stage. If a physical action is performed (self-rescue), a rescue is modeled in line 52 from a second stage of hypothermia. The command models a transition to a fatal stage in line 56. In Stage 3, the command in line 61 models the transition to state S_5 , representing a rescue. A full recovery, where the human returns to the normal state, is modeled in line 67.

5. Communication System Modeling in Galileo Satellites

Figure 3 depicts the SAR/Galileo Services, illustrating the organizational structure of rescue services. The system encompasses multiple components that relay beacon signals from distressed users to rescue authorities. This section presents the system parameters and the formal response mode model.

Listing 1: PRISM Model for Hypothermia Stages

```

1  ctmc
2  // System state constants
3  const int Normal = 5;
4  const int Hypothermia_STAGE_I = 4;
5  const int Hypothermia_STAGE_II = 3;
6  const int Hypothermia_STAGE_III = 2;
7  const int Hypothermia_STAGE_IV = 1;
8  const int S5 = 0; // rescued state
9
10 // Degradation rates (Mean Time Between Stages)
11 const double  $\lambda_{h1}$ ; // Normal -> Stage I
12 const double  $\lambda_{h2}$ ; // Stage I -> Stage II
13 const double  $\lambda_{h3}$ ; // Stage II -> Stage III
14 const double  $\lambda_{h4}$ ; // Stage III -> Stage IV
15
16 // Movement/recovery rates (based on MTS)
17 const double  $\mu_{h1}$ ; // Recovery at Stage I
18 const double  $\mu_{h2}$ ; // Recovery at Stage II
19 const double  $\mu_{h3}$ ; // Recovery at Stage III
20
21 module Hypothermia_Human_Degradation
22   S : [0..5] init Normal;
23   // [to_stage1] is the action name for this transition.
24   // S=Normal: The transition occurs when the human is in the 'Normal' state.
25   // ->  $\lambda_{h1}$  : (S'=Hypothermia_STAGE_I);
26   // -  $\lambda_{h1}$ : This is the rate parameter of an exponential distribution
27   //         that governs the time spent in the 'Normal' state before transitioning
28   //         to Hypothermia Stage I.
29   // - (S'=Hypothermia_STAGE_I): The human transitions to Hypothermia Stage I.
30   [to_stage1] S=Normal ->  $\lambda_{h1}$  : (S'=Hypothermia_STAGE_I);
31
32   // [to_stage2] is the action name.
33   // S=Hypothermia_STAGE_I: The transition occurs from Hypothermia Stage I.
34   // ->  $\lambda_{h2}$  : (S'=Hypothermia_STAGE_II);
35   // -  $\lambda_{h2}$ : Rate of progression from Stage I to Stage II.
36   // - (S'=Hypothermia_STAGE_II): The human transitions to Hypothermia Stage II.
37   [to_stage2] S=Hypothermia_STAGE_I ->  $\lambda_{h2}$  : (S'=Hypothermia_STAGE_II);
38
39   // [rescue1] is the action name for a successful rescue.
40   // S=Hypothermia_STAGE_I: The rescue attempt occurs when the human is in Hypothermia Stage I
41   // ->  $\mu_{h1}$  : (S'=S5);
42   // -  $\mu_{h1}$ : This is the rate of successful rescue from Stage I.
43   //         It runs in parallel with the 'to_stage2' transition, meaning
44   //         the human can either get worse or be rescued while in Stage I.
45   // - (S'=S5): The human transitions to state S5, which is likely a 'Rescued' or 'Recovery
46   //         ' state.
47   [rescue1] S=Hypothermia_STAGE_I ->  $\mu_{h1}$  : (S'=S5);
48
49   // Similar logic to [to_stage2], but for the next stage.
50   [to_stage3] S=Hypothermia_STAGE_II ->  $\lambda_{h3}$  : (S'=Hypothermia_STAGE_III);
51
52   // Similar logic to [rescue1], but for Stage II.
53   [rescue2] S=Hypothermia_STAGE_II ->  $\mu_{h3}$  : (S'=S5);
54
55   // Similar logic to previous progression steps.
56   // Stage IV is often considered near-fatal
57   [to_stage4] S=Hypothermia_STAGE_III ->  $\lambda_{h4}$  : (S'=Hypothermia_STAGE_IV);
58
59   // Similar logic to previous rescue steps.
60   //  $\mu_{h3}$  would reflect the extreme difficulty of rescue
61   // from severe hypothermia.
62   [rescue3] S=Hypothermia_STAGE_III ->  $\mu_{h3}$  : (S'=S5);
63
64   // State transition: Full recovery from the 'Rescued' state
65   // [full_recovery] is the action name.
66   // S=5: It refers to the numerical state 5, which is assumed to be 'S5'.
67   // - (S'=Normal): The human returns to the initial 'Normal' state, signifying full
68   //         recovery.
69   [full_recovery] S=S5 -> (S'=Normal);
70
71 endmodule

```

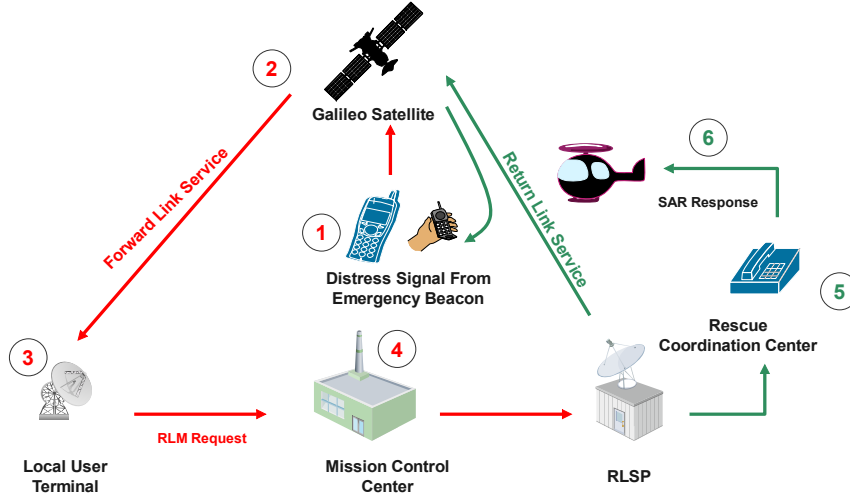


Figure 3: SAR/Galileo Services [4, 5].

5.1. The SAR system model

COSPAS-SARSAT (C/S) is an international satellite-based Search and Rescue (SAR) distress alerting system established in 1979 by the USA, Canada, France, and the former USSR [48]. (See Figure 3). The full documentation is available in [49]. The C/S system comprises:

1. Beacons are 406 MHz radio transmitting devices employed in various applications. These include Emergency Position Indicating Radio Beacons (EPIRBs) and Ship Security Alert Systems (SSAS) for maritime use, Emergency Locator Transmitters (ELTs) and ELT Distress Tracking for aviation, and Personal Locator Beacons (PLBs) for individual use.
2. The space segment includes satellites operating in low Earth orbit, geostationary orbit, and medium Earth orbit, responsible for processing signals transmitted by beacons.
3. A Service Ground Segment (SGS) consists of a geographically distributed set of receiving ground stations known as Local User Terminals (LUTs). This network provides ground segment coverage, allowing the tracking of satellites and the generation of independent location estimates for user beacons.
4. Mission Control Centers (MCC) are crucial in distributing C/S distress alerts globally and configuring the alerts for optimal response.
5. The Sar/Galileo Ground Segment consists of five Reference Beacons (REFBE). These reference beacons distributed across the European coverage Area are used to monitor the performance of the SAR/Galileo Service.

The Galileo Search and Rescue (SAR) Forward link service is capable of receiving signals emitted by C/S compatible 406 MHz distress beacons (Forward Link Alert Message, as FLAM) and relaying this information to a ground segment network, known as the Local User Terminal (LUT), which consists of geographically distributed facilities deployed worldwide. The Return Link Service (RLS) enables the relay of data (Return Link Messages, or RLMs) back to the originating beacon. A primary function of the RLS is to provide the end-user of a distress beacon with automatic acknowledgment, confirming the detection of the alert and the determination of their location by the Search and Rescue (C/S) system.

Upon estimating the beacon's location, the Mission Control Center (MCC) issues an RLM_Request. This request, covering the beacon's confirmed position, is transmitted to a backup MCC (Spanish or French). Subsequently, the RLS generates an RLM Transmission Request (RLMR) based on the original FLAM.

The Galileo Core infrastructure processes the RLMR and uplinks the RLM to appropriate Galileo satellites, which are then broadcast to the originating beacon.

Once the beacon receives the RLM, sets a receipt status flag within FLAM. This status flag is then transmitted to the RLSP via the C/S MCC. After the RLSP acknowledges the receipt of the RLM, the Galileo system stops sending further RLMs to the beacon. However, if no acknowledgment of RLM reception is received within 24 hours of the initial RLM request, the Galileo system will continue to transmit RLMs to the beacon.

5.2. Operational capabilities characteristics

According to [4] and [5], Galileo satellites demonstrate a reliability exceeding 88% over a 12-year lifespan. The work in [6] reports that the availability of healthy signals from Galileo is 99.22%, with a recovery time of less than 15 hours. This emphasizes the importance of ensuring timely service recovery. Detection performance, a crucial aspect of the system, refers to the probability of successfully detecting 406 MHz beacon transmissions within the SAR/Galileo coverage area and receiving a valid beacon message at the SAR/Galileo LUT Facilities. The summary of satellite parameters is portrayed in Table 4. To calculate the failure rate λ from satellite reliability (R) assuming an exponential distribution, we can use the following formula:

$$R(t) = e^{-\lambda t} \quad (4)$$

Solving for the failure rate λ :

$$\lambda = -\frac{\ln(R)}{t}$$

Given, reliability over 12 years: $R = 0.88$ and Mission time: $t = 12 \times 365 \times 24 \times 60 = 6,307,200$ minutes, the failure rate of satellite system λ_s .

$$\lambda_s = -\frac{\ln(0.88)}{6,307,200} = \frac{0.1278}{6,307,200} \approx 2.03 \times 10^{-8} \text{ failures/min}$$

Knowing that the availability of the healthy signal is $A = 99.22\%$, the failure rate λ and the recovery rate μ can be computed using the availability formula:

$$A = \frac{\mu}{\lambda + \mu} \implies \lambda = \mu \left(\frac{1 - A}{A} \right) \quad (5)$$

Assuming the mean service recovery time for the healthy signal is $T_{\text{rec,HS}} = 15 \text{ hours} = 900 \text{ minutes}$, the recovery rate is:

$$\mu_{\text{HS}} = \frac{1}{T_{\text{rec,HS}}} = \frac{1}{900} \approx 1.11 \times 10^{-3} \text{ per minute}$$

Then, using the availability to compute the failure rate:

$$\lambda_{\text{HS}} = \mu_{\text{HS}} \left(\frac{1 - 0.9922}{0.9922} \right) = 1.11 \times 10^{-3} \times \left(\frac{0.0078}{0.9922} \right) \approx 8.7 \times 10^{-6} \text{ per minute}$$

Thus, the computed rates for the healthy signal are:

$$\lambda_{\text{HS}} \approx 8.7 \times 10^{-6} \quad \text{and} \quad \mu_{\text{HS}} \approx 1.11 \times 10^{-3}$$

Three service states are defined for the SAR Forward link service in the ECA (European Coverage Area): Nominal, Degraded, and Severely Degraded. Nominal indicates normal operation. Degraded is characterized by either non-operational status for less than 24 hours continuously or less than 48 hours cumulatively over a calendar month, or by losing communication with one or two ground segments (LUTs) for more than one day, or only 1 REFBE is in nominal status, or no REFBE is operational for less than five continuous days.

Parameter	Value	Rate Type	Computed Rate (1/min)
Satellite Reliability over 12 years	88%	Failure Rate λ_s	2.03×10^{-8}
Satellite Recovery over 12 years	88%	Recovery Rate μ_s	4.04×10^{-6}
Healthy Signal Availability	99.22%	Failure Rate λ_{HS}	8.7×10^{-6}
Service Recovery Time ($T_{\text{rec,HS}}$)	15 hours	Recovery Rate μ_{HS}	1.11×10^{-3}

Table 4: Extracted Parameters and Computed Rates for Galileo SAR System

Severely Degraded occurs when the SGS is non-operational for more than 24 hours continuously or more than 48 hours cumulatively over a calendar month, or when communication is lost with all three LUTs and MCCs for more than four hours. According to [50], FLS availability is more than 99%. The summary of degradation rates is presented in Table 5.

If degradation happens on average after $MTTD$ minutes (Mean Time To Degrade), the degradation rate λ_d can be calculated using the exponential distribution formula:

$$\lambda_d = \frac{1}{MTTD} \quad (6)$$

Where λ_d is the degradation rate (in failures per minute), $MTTD$ is the mean time to degrade (in minutes). This assumption is based on the degradation process following an exponential distribution. In addition, the repair rate is calculated as follows:

$$\mu_r = \lambda_d \times \frac{A}{1 - A} \quad (7)$$

Where λ_d is the degradation (failure) rate, A is the availability target, and μ_r is the repair rate. We use Equations 6 and 7 to compute FLS and RLS degradation. In addition, λ_d and λ_{sd} model the degradation and severe degradation rates, whereas μ_d and μ_{sd} model the repair rates in case of degradation and severe degradation.

State	Condition Description	Threshold	Degradation Rate λ_d/λ_{sd}	Repair Rate μ_r/μ_{sr}
Nominal	Normal operation	—	—	—
Degraded	Non-operational for less than 24 hours continuously	< 24 h continuous	6.94×10^{-4}	1.39×10^{-3}
	Less than 48 hours cumulatively per month	< 48 h per month	3.47×10^{-4}	1.39×10^{-3}
	Communication loss with 1-2 LUTs for more than 1 day	> 1 day (LUTs)	6.94×10^{-4}	1.00×10^{-3}
	Only 1 REFBE is nominal	—	—	6.94×10^{-4}
	No REFBE operational for less than 5 days	< 5 days (REFBE)	1.39×10^{-5}	2.78×10^{-4}
Severely Degraded	Non-operational for more than 24 hours continuously	> 24 h continuous	6.94×10^{-4}	1.39×10^{-3}
	More than 48 hours cumulatively per month	> 48 h per month	3.47×10^{-4}	6.94×10^{-4}
	Communication loss with all 3 LUTs and MCCs for more than 4 hours	> 4 h (full comm loss)	4.17×10^{-3}	1.39×10^{-3}

Table 5: Forward Link Service Degradation states with condition-by-condition thresholds and corresponding degradation rates λ_d and repair rates μ_r (1/min).

Similarly, the RLS has three service states: Nominal, Degraded, and Severely Degraded. The system is considered “Degraded” if the RLSP is in degraded status or not operational for up to 7 cumulative hours within a calendar month or if RLM messages are delivered but not compliant with the latency MPL: The RLM Delivery Latency within 15 min was above or equal to 99.95% [4]) which means that the Failure Rate is 0.05%. “Severely Degraded” status applies when the RLSP is not operational for more than seven cumulative hours within a calendar month or if RLM messages are not delivered for more than 7 hours. Additionally, the given availability of the RLS is 95%. We provide a summary of the computation in Tble 6.

State	Condition Description	Threshold	Degradation Rate λ_d/λ_{sd}	Repair Rate μ_r/μ_{sr}
Nominal	Normal operation	—	—	—
Degraded	Non-operational for less than 7 hours continuously RLM compliance with MPL rate	< 7 h continuous	2.38×10^{-3}	4.52×10^{-2}
			8.33×10^{-6}	1.67×10^{-2}
Severely Degraded	Non-operational for more than 7 hours continuously	> 7 h continuous	2.08×10^{-3}	3.95×10^{-2}

Table 6: Degradation states with condition-by-condition thresholds and corresponding degradation rates λ_d and repair rates μ_r (1/min).

5.3. A formal model of satellite communication degradation

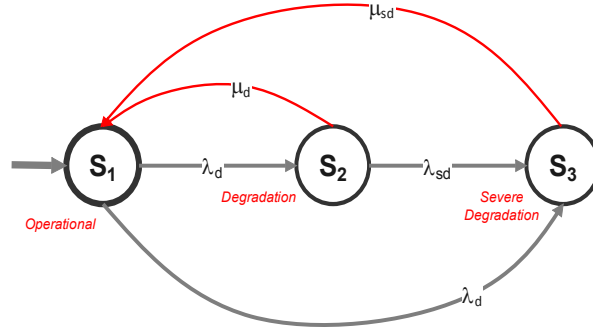


Figure 4: Model for SAR Services and Human Degradation.

The SAR system services (RLS and FLS) are modeled as a three-state Markov chain, illustrated in Figure 4. Its PRISM textual representation is provided in Listing 2. State S_1 (initial state) represents the fully operational system, while state S_2 indicates a faulty state requiring recovery. State S_3 signifies a severe degradation state where maintenance is required. Degradation transitions are indicated by black arrows, while recovery is shown with red arrows.

In the PRISM model, these system states are represented by an integer variable S , taking values from 0 to 2 (line 14). The assignment of degradation status relies on constant values defined in lines 3-5. In this model, λ_d and λ_{sd} represent the failure rates of SAR system services components (identified previously) in their degraded and severely degraded states, respectively, and μ_d and μ_{sd} indicates the repair rate in degraded and sever degradation mode. The values for these degradation rates were identified in previous sections.

The PRISM global variables are parameterizable in Listing 2 (lines 8-11). These variables are set according to data from the previous section, encompassing various sources of failure. This allows the model to be parameterized based on the specific failure source (e.g., degradation over 24/48 hours, communication loss with ECA, and status of REFBE).

Listing 2: The Service Status Operational Rates

```

1  ctmc
2  //System states
3  const int Operational = 2;
4  const int Degraded = 1;
5  const int Severely_Degraded = 0;
6
7  //System performance rates
8  const double  $\lambda_d$ ; // Failure rate (degradation) based on MTBF for Degraded state
9  const double  $\lambda_{sd}$ ; // Failure rate (degradation) based on MTBF for Severely Degraded state
10 const double  $\mu_d$ ; // Repair rate based on MTTR for Degraded state
11 const double  $\mu_{sd}$ ; // Repair rate based on MTTR for Severely Degraded state
12 //PRISM module
13 module Service_Degradation
14   S : [0..2] init Operational;
15   [degradation] S>0  $\rightarrow \lambda_d : (S' = \text{Degraded})$ ;
16   [sever_degradation] S>0  $\rightarrow \lambda_{sd} : (S' = \text{Severely\_Degraded})$ ;
17   [reset_degradation] S=Degraded  $\rightarrow \mu_d : (S' = \text{Operational})$ ;
18   [reset_sever_degradation] S=Severely_Degraded  $\rightarrow \mu_{sd} : (S' = \text{Operational})$ ;
19 endmodule

```

6. Quantitative Analysis using PRISM

In this section, we verify the performance of the satellite system in relation to its dedicated actions for transmitting rescue signals. First, we evaluate the reliability of the satellite system in the presence of different failures induced by communication reliability signals. Then, we incorporate human factors derived from hypothermia status, with the composed model providing accurate visibility into the quality of service and the status of humans in danger.

Experimental Setup. We encoded properties in CSL for the CTMC formalism and in rPATL for the CSG model formalism. PRISM model checker v4.8 performs probabilistic model checking, and PRISM-games model checker v3.0 [51] (based on PRISM 4.6) performs probabilistic verification on game models. These experiments were conducted on an Ubuntu 20 system equipped with 32GB RAM. Various engines are available to enhance performance for specific model structures, depending on the model complexity (see documentation [34]).

Artifacts. The source code used for the experiments detailed in this section and throughout the paper is publicly accessible on the website [52].

6.1. Galileo satellite system dependability

The PRISM tool can evaluate a wide range of dependability properties. In this work, we are particularly interested in assessing signal transmission performance, initially focusing on scenarios where the distressed individual maintains a good psychological status, and excluding human reliability in this first step.

6.1.1. Safety

The property “*Safety*” evaluates the satellite signal performance by calculating the probability of the satellite successfully collecting the signal from a beacon while facing various signal degradation, represented by the label “**System_Fully_Operational**”. To define such a label, we enhance the PRISM model with the following listing code, portrayed in Listing 3. This PRISM formula and its associated

label collaboratively define the precise condition for the entire satellite system to be considered fully operational. It models the satellite as a whole system functioning without any degradation. The formula, named `System_Fully_Operational_Up`, evaluates to true if and only if:

- The Satellite On-Orbit component (`SatteliteOnOrbit_s`) is in its Operational state.
- The Healthy Satellite Signal component (`HealthySatteliteSignal_s`) is in its Operational state.
- The FLS Satellite Signal component (`FLSSatteliteSignal_s`) is in its Operational state, indicating the Forward Link Signal is functioning optimally without any degradation.
- The RLS Satellite Signal component (`RLSSatteliteSignal_s`) is in its Operational state, meaning the Return Link Signal is also functioning optimally without any degradation.

This formula is a conjunction (logical AND) of the operational status of all its critical subsystems. If any one of these four critical components is not in its Operational state (i.e., it is in a Degraded or `Severely_Degraded` state), then the `System_Fully_Operational_Up` formula will evaluate to false. The label “`System_Fully_Operational`” provides a human-readable and convenient alias for the `System_Fully_Operational_Up` formula. Instead of rewriting the complex condition for the entire system’s full operational status in properties or other parts of the model, we can refer to “`System_Fully_Operational`”.

Listing 3: The Service Degradation Formula and Labels

```

1 //A formula for the entire "satellite system" being up (all components operational).
2 formula Overall_Satellite_System_Up =
3   (SatteliteOnOrbit_s = Operational) &
4   (HealthySatteliteSignal_s = Operational) &
5   (FLSSatteliteSignal_s = Operational) &
6   (RLSSatteliteSignal_s = Operational);
7
8 // Label for the overall satellite system being fully operational.
9 label "System_Fully_Operational" = Overall_Satellite_System_Up;
10
11 // Formula for the system being in a severely degraded mode.
12 // This occurs if either the FLS or RLS signal is in a Severely_Degraded state.
13 // Note: SatteliteOnOrbit and HealthySatteliteSignal modules do not have a Severely_Degraded
   state.
14 formula System_In_Severely_Degraded_Mode =
15   (FLSSatteliteSignal_s = Severely_Degraded) |
16   (RLSSatteliteSignal_s = Severely_Degraded);
17
18 // Formula for the system being in a general degraded mode.
19 // This means the system is NOT fully operational, AND NOT in a severely degraded mode.
20 formula System_In_Degraded_Mode =
21   !Overall_Satellite_System_Up &           // System is NOT fully operational
22   !System_In_Severely_Degraded_Mode;      // AND NOT in a severely degraded state
23
24 // Labels for the new formulas
25 label "System_Severely_Degraded" = System_In_Severely_Degraded_Mode;
26 label "System_Degraded" = System_In_Degraded_Mode;
27
28 // Rewards: Measures satellite system availability for rescue operations.
29 rewards "rescue_availability"
30   // Accumulate 1 unit of reward per unit time when the system is fully operational.
31   Overall_Satellite_System_Up : 1;
32 endrewards

```

The results indicate a 100% probability of the system transitioning from an operational state to a degraded state. The results of the PRISM analysis indicate a 100% probability that the satellite system, initially in its “`System_Fully_Operational`” state, will eventually transition to a non-fully-operational state ($\neg$ “`System_Fully_Operational`”). Thus, this leads us to investigate the role of degradation features on the signal transmitted through the verification of properties “*Degraded*” and “*Severely Degraded*”.

Property 1

$$P=? [G(\text{"System_Fully_Operational"} \implies F\neg\text{"System_Fully_Operational"})] \quad (\text{"Safety"})$$

6.1.2. Reliability

In this section, we verify the property *"Degraded"*. This property assesses whether, as time elapses up to k , the system transitions from a fully operational state to a general degraded state. Our analysis tracks explicitly the degradation path according to the scenario-based features presented in Table 5, allowing us to identify the primary factors contributing to the increasing degradation over time. To facilitate this, we enhance the model with two PRISM formulas, defined in Listing 3 (lines 14-22). The first formula, *System_In_Severely_Degraded_Mode*, expresses the state of severe degradation, triggered, which is true if either the Forward Link Signal (FLS) or Return Link Signal (RLS) is in a *Severely_Degraded* state. The second formula, *System_In_Degraded_Mode*, defines the general degraded state. This state is true when the overall satellite system is not fully operational ($\neg\text{Overall_Satellite_System_Up}$) and not in a severely degraded mode ($\neg\text{System_In_Severely_Degraded_Mode}$). This distinction enables us to precisely capture different levels of system impairment, extending beyond complete failure.

Property 2

$$P=? [(\text{"System_Fully_Operational"}) U^{\leq k} (\text{"System_Degraded"})] \quad (\text{"Degraded"})$$

In the remainder of the verification procedure, we focus only on the best-case and worst-case scenarios of reliability assessment, where the signal remains reliable over the long or short term, respectively, as defined by the parameter k over 30,000 minutes. Our goal is to assess the types of degradation that have positive or negative impacts.

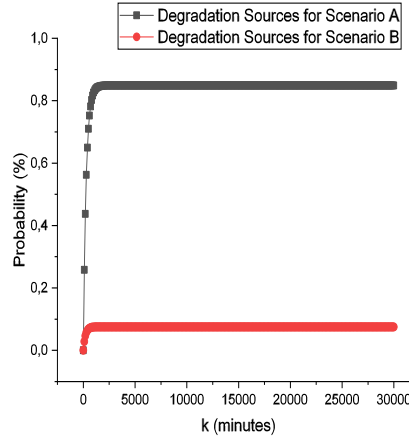


Figure 5: Verification of Property *"Degraded"* for Scenarios A and B.

Figure 5 illustrates that satellite signal transmission remains reliable until degradation is observed. We performed verification against property *"Degraded"* for all combinations of degradation to assess the best and worst cases. This highlights the most reliable signal transmission in both scenarios. **Scenario A** demonstrates that the signal remains reliable with FLS and RLS degradation stemming from less than 24 hours of continuous non-operational status and less than 7 hours of continuous non-operational status, respectively. However, severe degradation from FLS stems from less than 48 hours of cumulative monthly downtime, resulting in a reliability of 84%. In the worst-case scenario (**Scenario B**), reliability remains at 7.5% due to FLS and RLS degradation stemming from less than 48 hours of cumulative monthly downtime. Severe FLS degradation is also captured due to communication loss (1-2 LUTs) lasting more than 1 day.

Moreover, we verify Property “*Severely Degraded*”, which assesses the system’s transition from a fully operational state to a severely degraded state over time, up to a specified time k . Similar to Property “*Degraded*”, our analysis explicitly tracks the degradation path based on the scenario-driven features detailed in Table 5. This enables us to identify the primary factors contributing to the increasing severity of degradation over time. For this analysis, we use the PRISM formula **System_In_Severely_Degraded_Mode**, which is true if either the Forward Link Signal (FLS) or Return Link Signal (RLS) enters a **Severely_Degraded** state. This formula is conveniently represented by the label “**System_Severely_Degraded**”.

Property 3

$$P = ? [(\text{“System_Fully_Operational”}) \cup^{\leq k} (\text{“System_Severely_Degraded”})] \quad (\text{“Severely Degraded”})$$

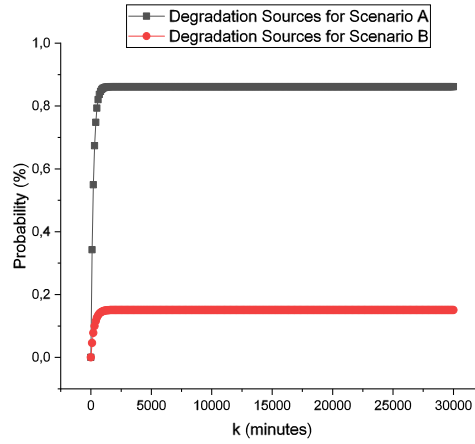


Figure 6: Verification of Property “*Severely Degraded*” for Scenarios A and B.

Figure 6 illustrates the analysis over 30,000 minutes of execution time for signal transmission. The assessment was performed across all combinations of degradation to identify the best and worst combinations that lead to reduced signal performance. In this figure, we display only the performance for degradation in **Scenario A**, with reliability that remains 86% and includes FLS degradation and severe degradation due to communication loss (1-2 LUTs) for more than 1 day. This is in addition to RLS degradation stemming from RLM non-compliance (considering the original severe degradation of RLS for more than seven continuous hours). The worst-case scenario, **Scenario B**, with reliability 15% includes degradation stemming from less than 24 hours of continuous non-operational status and severe degradation due to more than 48 hours of cumulative monthly downtime. It also includes degradation of RLS observed for less than 7 continuous non-operational hours.

6.1.3. Operational rescue availability

Property 4

$$R\{\text{“rescue_availability”}\} = ? [C^{\leq k}] / k \quad (\text{“Mission Rescue Availability”})$$

In Property “*Mission Rescue Availability*”, we evaluate the interval availability (or expected accumulated uptime) of the satellite system for performing a rescue operation. This property quantifies the total time, over a specified interval k , that the satellite system is capable of executing a complete rescue sequence. Such a sequence involves the satellite successfully collecting a signal from a beacon, transmitting it through FLS services, initiating an RLM request, alerting a rescue coordination center, and ultimately ensuring

the correct provision of RLS services. The availability is measured by the reward structure as portrayed in Listing 3 lines 29-32; this reward structure accumulates 1 unit of reward for every unit of time the `Overall_Satellite_System_Up` formula evaluates to true. This ensures that the accumulated reward directly reflects the duration for which the satellite system is in its fully operational state.

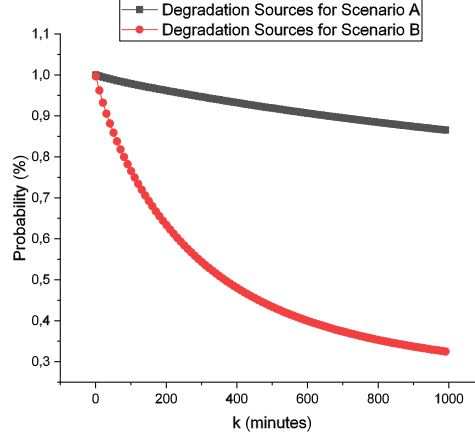


Figure 7: Verification of Property “*Mission Rescue Availability*”.

Figure 7 illustrates the availability of the communication signal for the best and worst-case scenarios involving combinations of FLS and RLS signal degradation. In **Scenario A**, the FLS and RLS degradation sources represent less than 24 hours of continuous non-operational time and less than 7 hours of continuous non-operational time, respectively. However, severe FLS degradation is due to more than 48 hours of cumulative monthly downtime, causing the probability to decrease from 99% to 86%. The worst-case scenario, **Scenario B**, indicates that FLS and RLS degradation originates from communication loss (1-2 LUTs) lasting more than 1 day and exceeding 24 hours of continuous non-operational time. Additionally, severe FLS degradation is caused by communication loss (all 3 LUTs/MCCs) for more than 4 hours.

6.2. Human-in-the-Loop satellite system dependability

The previous results do not provide evidence that a rescue operation ensures the survival of the individual. Indeed, while the factors of system degradation portrayed earlier offer insights into such rescue missions, we do not yet fully understand the impact of the human factor, particularly **hypothermia** (see Section 4). This lack of understanding can lead to a scenario where, despite the satellite system being operationally available, it cannot effectively manage the urgent situation of a human at sea seeking rescue.

Modeling such a configuration, which incorporates human factors and satellite system features, can provide clear insights, offering mission operators accurate reasoning regarding mission reliability, system degradation, and human status. To achieve this, the PRISM models for human **hypothermia** (see Section 4) status and satellite status are combined into a single PRISM model. As a result, the PRISM model’s formulas and labels are rewritten to reflect this new configuration.

So, the model is augmented by human status related to **hypothermia**; the PRISM code for this module is available in Section 4. We reuse the same notation from that section for this purpose. The additional formulas, labels, and rewards integrating human status are detailed in Listing 4. The label on line 5 (“`System_Severely_Degraded_With_Unrecovered_Human`”) refers to the severe satellite degradation state with the human at a specific hypothermia stage. Similarly, the label on line 8 (“`System_Degraded_With_Unrecovered_Human`”) defines the degraded system status along with the human’s non-recovered hypothermia status and the overall degradation of the satellite system. Finally, the reward structure in lines 14-21 (“`rescue_metrics`”) expresses comprehensive rescue metrics. This is an extension of the previous reward structure, which also incorporates a reference to the hypothermia stage.

Listing 4: PRISM Definitions for System and Human Status

```

1 // Labels for combined system and human states.
2
3 // Defines a critical state where the satellite system is in a severely degraded mode
4 // AND the human's hypothermia has reached a specific STAGE.
5 label "System_Severely_Degraded_With_Unrecovered_Human" = System_In_Severely_Degraded_Mode & (s
  = STAGE);
6
7 // Defines a state where the satellite system is in a general degraded mode
8 // AND the human's hypothermia has reached a specific STAGE.
9 label "System_Degraded_With_Unrecovered_Human" = System_In_Degraded_Mode & (s = STAGE);
10
11 --- Rewards : rescue_metrics ---
12 // This reward module quantifies specific metrics relevant to system availability
13 // combined with the human's hypothermia status during a rescue scenario.
14 rewards "rescue_metrics"
15   // State-based reward: Accumulates 1 unit of reward per unit of time
16   // when the satellite system is fully operational AND the human's hypothermia
17   // is at the specified STAGE. This measures the total time the system is optimally
18   // functional while the human is in this particular critical state, which might
19   // represent a specific window for intervention or assessment.
20   Overall_Satellite_System_Up & (s = STAGE) : 1;
21 endrewards

```

6.2.1. Reliability

With the incorporation of the new labels and rewards, we can now express more refined properties for the combined model that integrates both human status and satellite system degradation levels. We refine Property “*Degraded*” to express Property “*Degraded With Unrecovered Human*”, which explicitly includes the human’s unrecovered status. Additionally, we refine Property “*Severely Degraded*” to express Property “*Severely Degraded With Unrecovered Human*”, which incorporates the human’s unrecovered status alongside the severe degradation of the SAR system.

Property 5

$$P = ? [(\text{“System_Fully_Operational”}) \quad (\text{“Degraded With Unrecovered Human”}) \\ U^{\leq k} (\text{“System_Degraded_With_Unrecovered_Human”})]$$

The results of verifying Property “*Degraded With Unrecovered Human*” during the different stages of hypothermia (k takes the overall hypothermia cycle - 60 minutes) in Figure 8 show that human factors significantly impact Scenario A, compromising mission reliability, as human degradation due to hypothermia is highlighted. Specifically, while the mission success probability in the best-case of Scenario A (without explicit human impact) showed an increasing trend over 57 minutes, reaching 15% despite system degradation, the inclusion of human factors imposes a significantly lower, stable reliability: 0.3% for hypothermia at Stage 1, and only 0.05% for hypothermia at Stage 4.

Property 6

$$P = ? [(\text{“System_Fully_Operational”}) \quad (\text{“Severely Degraded With Unrecovered Human”}) \\ U^{\leq k} (\text{“System_Severely_Degraded_With_Unrecovered_Human”})]$$

Similarly, the results of verifying Property “*Severely Degraded With Unrecovered Human*” are portrayed in Figure 9 and show that, in Scenario A (which includes FLS degradation, severe degradation due to

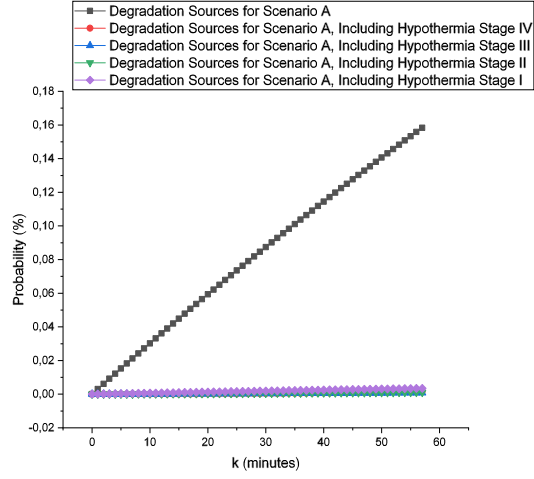


Figure 8: Verification of Property “*Degraded With Unrecovered Human*”.

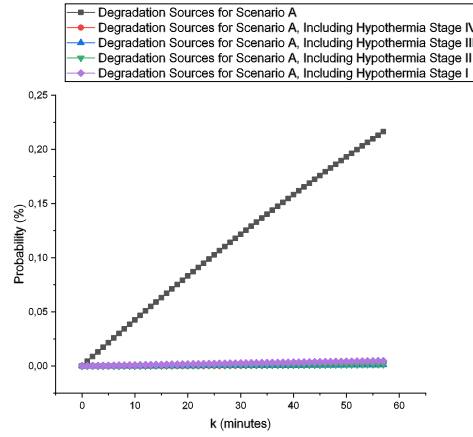


Figure 9: Verification of Property “*Severely Degraded With Unrecovered Human*”.

communication loss (1-2 LUTs), and RLS degradation for more than 1 day), the human factor, such as hypothermia, has a significant impact on the reliability of the rescue mission.

Consequently, for both types of satellite signal degradation, it is evident that performing efficient rescue interventions at the first stage of hypothermia is crucial, as human survival is highly probable at this point. In our study, we have conducted a mathematical proof to validate these assumptions.

6.2.2. Operational rescue availability

Property 7

$$R\{\text{"rescue_metrics"}\} = ?[C^{\leq k}]/k \quad (\text{"Rescue with Human Status"})$$

In Property *"Rescue with Human Status"*, we refine the property expressed in Property *"Mission Rescue Availability"* to include the status of the human under hypothermia, as its structure is defined in Listing 4. Thus, this property will now consider not only the satellite's status but also the human's condition (hypothermia) during rescue operations.

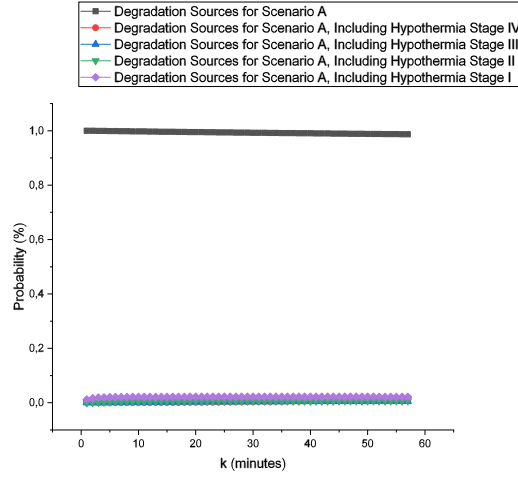


Figure 10: Verification of Property *"Rescue with Human Status"*.

The results of verifying Property *"Rescue with Human Status"*, portrayed in Figure 10, show a significant discrepancy in mission availability in Scenario A when comparing cases independent of human hypothermia status versus those that include it. **Scenario A** itself includes FLS degradation sources representing less than 24 hours of continuous non-operational time and RLS degradation sources representing less than 7 hours of continuous non-operational time, respectively. While mission availability is close to 90% when human hypothermia is not considered, including the human hypothermia status significantly decreases mission availability to approximately 2% for the first stage of hypothermia, and to 0.5% for hypothermia Stages 2 and 3. Interestingly, Stage 4 shows an increasing probability, which is likely due to the human signal becoming unavailable.

6.3. Discussion

6.3.1. Lessons Learned from Human-in-the-Loop

In this study, we investigate the reliability and availability of the rescue mission within the Galileo SAR system. We performed verification on these properties and investigated which degradation scenarios impact mission efficacy. Additionally, we enhanced the model by incorporating human degradation due to hypothermia, using hypothermia stages collected from clinical observations. Upon combining the human factor, mission reliability decreased. While this outcome is consistent with our assumptions, it has been mathematically proven, and we believe this represents the first such verification performed. This study is

open to considering multiple combined situations, as humans are not only sensitive to hypothermia but also to various other *factors* that can decrease their physiological and psychological performance.

6.3.2. From Markov model to game-theoretical model

The assessment performed in the previous section is based on CTMC models, which enable us to investigate the performance of satellite systems over time, as well as the status of human degradation. This approach is primarily based on models suitable for a wide range of use cases, supported by the PRISM probabilistic model checker.

However, humans are inherently combative by nature; when sending signals to satellite systems, they possess capabilities to self-rescue by maintaining self-stability and continually striving to stay alive. In a human’s consciousness, this active struggle can be represented by the number of rounds needed to overcome the situation, driven by their instinct for survival. Mathematically, this dynamic can be expressed as a game. To do so, we gather the values collected from previous experiments to model a game model in CSG as portrayed in Listing 5.

Listing 5 portrays three modules that model the human body’s hypothermia status, the satellite system’s status, and a synchronized or recorder module that orchestrates the “game” between human and satellite systems. Most of the constant variables (lines 5-18) used in the model are detailed in the artifact section. Readers can download the model and experiment with variations of human probabilities.

The “**Hypothermia_Human_Degradation**” module (lines 20-30) illustrates the different human body states as described earlier in Section 4.1. This module consists of three commands; the situation of hypothermia in Stage 4 is omitted as it is not considered at this level of analysis. The Satellite model’s behavior (lines 32-40) is based on reliability computed from previous experiments, which show 86% and 87% reliability in the degradation and severe degradation modes, respectively, for the best-case scenario (Scenario A), independent of human degradation. Meanwhile, the RescueOperation module (lines 42-55) performs synchronization between hypothermia stages based on parameters computed in Section 6.2, specifically for two degradation cases.

The model is also enhanced by incorporating the concept of “*number of rounds*”, which is further detailed in the artifacts section [52] and described in detail in [17] and in [53, 16]. This parameter quantifies the number of times a human attempts to maintain self-stability in the face of challenging cold situations. It is vital for us to calculate how capable human are of handling this situation and to develop a proper measure of their instinct for survival. Therefore, we verified Property “*Survival Attempts*” in both degradation and severe degradation modes.

Property 8

$\ll p1, p2 \gg \text{Pmax} = ?[F((\text{rescue} = 1) \& \text{rounds} \leq k)]$ (“*Survival Attempts*”)

Property “*Survival Attempts*” assesses that the probability of rescue reaches 1 (as defined by the variable in Listing 5). Here, rescue is realized with a previously computed probability, and the “**rounds**” represent the efficiency with which the mission may succeed, reflecting concepts quantified in CTMC models.

The results of such computations are portrayed in Figure 11. It shows that the delta (difference) is **approximately 2%** compared to the CTMC results, a discrepancy which may be significant for both mission success and human survival. However, we observe that the model shows high reliability even in cases of severe degradation. This is likely because the input parameters for the model considered severe degradation occurring after a significant amount of time, rather than degradation that may arise earlier and be caused by multiple factors.

6.4. Threats to validity

The modeled system incorporates parameters gathered from available online documentation and websites. These models could be further enhanced by integrating additional features, such as satellite movement parameters, to achieve a more accurate representation of Search and Rescue (SAR) systems. However, this paper includes a limited set of these parameters due to the current capabilities of the PRISM model checker and its inherent limitations concerning the state explosion problem if more modules were to be implemented.

Listing 5: PRISM Definitions for CSG model

```

1  csg
2  player p1 Hypothermia_Human_Degradation endplayer
3  player p2 Sattelite endplayer
4  // System state constants for human hypothermia stages and intervention outcomes and satellite
   operational modes.
5  ///..... Variable definitions are provided in the artifacts section.
6
7  // Probabilities of degrading to the next hypothermia stage per discrete time step.
8  // These are transition probabilities, indicating the likelihood of progression in one step.
9  const double p_h1=0.93; // Probability of degrading human from Normal to Hypothermia_STAGE_I.
10 const double p_h2=0.75; // Probability of degrading human from Hypothermia_STAGE_I to
   Hypothermia_STAGE_II.
11 const double p_h3=0.07; // Probability of degrading human from Hypothermia_STAGE_II to
   Hypothermia_STAGE_III.
12
13 // Satellite reliability: Probability of the satellite *remaining operational* per time step.
14 const double r=0.86; // Probability that the satellite does NOT degrade in a given time step.
15
16 const double e_h1; //Rescue operation winning probability in case of hypotermia stage 1
17 const double e_h2; //Rescue operation winning probability in case of hypotermia stage 2
18 const double e_h3; //Rescue operation winning probability in case of hypotermia stage 3
19
20 module Hypothermia_Human_Degradation
21
22     // State variable 's' representing the current hypothermia stage of the human.
23     // Initialized to 'Normal' (state 5), indicating a healthy starting condition.
24     s : [0..5] init Hypothermia_STAGE_I;
25
26     // Degradation transitions for the human's hypothermia.
27     [degrade_to_h1] s=Hypothermia_STAGE_I -> (1-p_h1) : (s'=Hypothermia_STAGE_II) + (p_h1) :
   (s'=Hypothermia_STAGE_I);
28     [degrade_to_h2] s=Hypothermia_STAGE_II -> (1-p_h2) : (s'=Hypothermia_STAGE_III) + (p_h2) :
   (s'=Hypothermia_STAGE_II);
29     [degrade_to_h3] s=Hypothermia_STAGE_III -> (1-p_h3) : (s'=Hypothermia_STAGE_IV) + (p_h3) :
   (s'=Hypothermia_STAGE_III);
30 endmodule
31
32 module Sattelite
33     // State variable 'Sattelite_s' representing the satellite's operational status.
34     // Initialized to 'Satellite_Operational' (state 0), indicating it starts functional.
35     Sattelite_s : [0..1] init Satellite_Operational;
36     // With probability (1-r), the satellite degrades to 'Satellite_Failed'.
37     // With probability 'r' (its reliability), it remains 'Satellite_Operational'.
38     [degrade_sattelite] Sattelite_s=Satellite_Operational -> (1-r) : (Sattelite_s'=
   Sattelite_Failed) + r : (Sattelite_s'=Satellite_Operational);
39     [repair] Sattelite_s=Satellite_Failed -> (Sattelite_s'=Satellite_Operational);
40 endmodule
41
42 module RescueOperation
43     // State variable to indicate the outcome of a rescue attempt (0: failure, 1: success).
44     rescue : [0..1] init 0;
45
46     // These transitions define the outcome of a rescue attempt
47     // The module takes this transition ONLY if both a human degradation event
48     // AND a satellite degradation event (degrade_sattelite) occur simultaneously.
49     // Rescue outcome when human degrades to Stage I AND satellite also attempts to degrade (but
   is operational).
50     [degrade_to_h1,degrade_sattelite] s=Hypothermia_STAGE_I & Sattelite_s=Satellite_Operational ->
   e_h1:(rescue'=1)+(1-e_h1):(rescue'=0);
51     // Rescue outcome when human degrades to Stage II AND satellite also attempts to degrade (but
   is operational).
52     [degrade_to_h2,degrade_sattelite] s=Hypothermia_STAGE_II & Sattelite_s=Satellite_Operational
   -> e_h2:(rescue'=1)+(1-e_h2):(rescue'=0);
53     // Rescue outcome when human degrades to Stage III AND satellite also attempts to degrade (but
   is operational).
54     [degrade_to_h3,degrade_sattelite] s=Hypothermia_STAGE_III & Sattelite_s=Satellite_Operational
   -> e_h3:(rescue'=1)+(1-e_h3):(rescue'=0);
55 endmodule

```

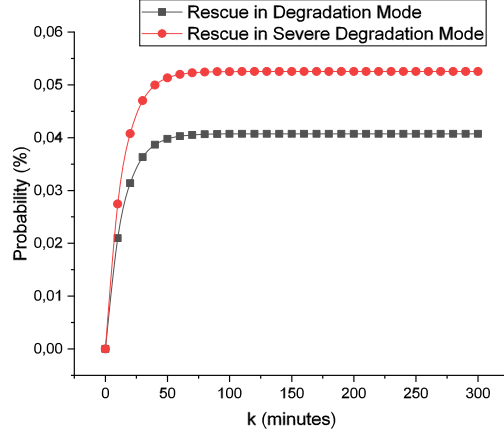


Figure 11: Verification of Property “*Survival Attempts*”.

For instance, only human hypothermia has been modeled, with its stages derived from the variability in the physiological and psychological status of the human body. While some studies provide parameters related to the time between different hypothermia stages, these often correlate with factors such as *genetics* and geographical localization. Furthermore, in other contexts, these dynamics are influenced by *cultural dependencies*, such as experienced situations or learned behaviors, which contribute to a person developing a skilled profile in such challenging circumstances.

7. Conclusion

This paper presents an approach based on CTMCs and CSGs to model the communication services of the SAR/Galileo system. The captured model incorporates multiple degradation scenarios related to the observed and monitored communication between satellite systems and ground stations. We have used the PRISM model checker for quantitative analysis, considering dependability parameters and the evolving status of the distressed person regarding hypothermia.

The analysis assesses which degradation source contributes most significantly to system failures and impacts SAR performance, referred to as Scanrio A. Our results demonstrate that combined degradation of FLS and RLS signals primarily contribute to failures. Furthermore, an evaluation was conducted to assess the human’s capability to react to a crisis, considering physiological and psychological variability (resulting from hypothermia) in conjunction with the system status. The findings indicate that the human status influences the rescue mission. This study performs a mathematical verification rather than relying on experienced assumptions, which represents a first in its application. While most analysts rely on experienced decision-makers, our approach leverages mathematical proofs that prioritize the automation of decision-making.

We have conducted a comparison with game-theoretical models that portray a minor significance when the human instinct for survival is factored in, as the individual will attempt to stabilize themselves to facilitate rescue after multiple rounds. From a philosophical perspective, this depends on multiple criteria for assessing which rounds best reflect human capabilities. These criteria can include genetics or cultural background, as individuals are raised with varying degrees of combative survival behavior.

Future work will include a comparative analysis of various factors obtained from clinical sources and human physiological characteristics. We recognize that each individual possesses unique characteristics that cannot be universally generalized; however, the model itself can nonetheless be parameterized to accommodate this inherent variability. Finally, we will undertake a comprehensive survey of parameters to show the relation between humans and cyber-physical systems, in order to contribute towards the development of truly socio-cyber-physical systems.

Acknowledgements

The research leading to the presented results was conducted within the research profile of **Human-Centric Collaborative Architectural Decision-Making for Secure System Design** (HERMES-Design¹) supported by Institut de Cybersécurité d'Occitanie (ICO).

The parameters related to hypothermia are derived from clinical literature sources. It is essential to acknowledge that these values can vary considerably between regions and individuals. Consequently, this study is not intended to provide medical guidance or serve as a reference for rescue operations; its sole purpose is to identify and model the complex relationship between human behavior and cyber-physical systems within a theoretical framework.

References

- [1] J. A. Fraire, S. Henn, G. Stock, R. Ohs, H. Hermanns, F. Walter, L. Van Broock, G. Ruffini, F. Machado, P. Serratti, J. Relloso, Quantitative analysis of segmented satellite network architectures: A maritime surveillance case study, *Computer Networks* 255 (2024) 110874. URL: <https://www.sciencedirect.com/science/article/pii/S1389128624007060>. doi:<https://doi.org/10.1016/j.comnet.2024.110874>.
- [2] P. Norris, Developments in high resolution imaging satellites for the military, *Space Policy* 27 (2011) 44–47. URL: <https://www.sciencedirect.com/science/article/pii/S0265964610001098>. doi:<https://doi.org/10.1016/j.spacepol.2010.12.003>.
- [3] The Economist, A New Constellation of Space Tie-Ups, 2024. URL: <https://impact.economist.com/projects/a-new-constellation-of-space-tie-ups/>, the Economist Impact.
- [4] E. S. A. (ESA), Galileo performances, https://gssc.esa.int/navipedia/index.php/Galileo_Performances, -. Accessed in January 2025.
- [5] E. U. A. for the Space Programme (EUSPA), Galileo open service system definition document (sdd) v1.3, https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo-OS-SDD_v1.3.pdf, Accessed in January 2025.
- [6] E. U. A. for the Space Programme (EUSPA), Galileo open service (os) performance reports, <https://www.gsc-europa.eu/electronic-library/performance-reports/galileo-open-service-os>, Accessed in January 2025.
- [7] C. Baier, J.-P. Katoen, Principles of model checking, The MIT Press, 2008. OCLC: ocn171152628.
- [8] M. Kwiatkowska, G. Norman, D. Parker, PRISM 4.0: Verification of probabilistic real-time systems, in: Proc. 23rd International Conference on Computer Aided Verification (CAV'11), volume 6806 of *LNCS*, Springer, 2011, pp. 585–591.
- [9] V. Forejt, M. Kwiatkowska, G. Norman, D. Parker, Automated verification techniques for probabilistic systems, in: Formal Methods for Eternal Networked Software Systems (SFM'11), volume 6659 of *LNCS*, Springer, 2011, pp. 53–113.
- [10] M. Kwiatkowska, G. Norman, D. Parker, Stochastic model checking, in: M. Bernardo, J. Hillston (Eds.), Formal Methods for the Design of Computer, Communication and Software Systems: Performance Evaluation (SFM'07), volume 4486 of *Lecture Notes in Computer Science*, Springer, 2007, pp. 220–270. doi:[10.1007/978-3-540-72522-0_6](https://doi.org/10.1007/978-3-540-72522-0_6).
- [11] M. Kwiatkowska, G. Norman, D. Parker, G. Santos, Automatic verification of concurrent stochastic systems, *Formal Methods Syst. Des.* 58 (2021) 188–250. URL: <https://doi.org/10.1007/s10703-020-00356-y>.
- [12] PRISM - Bibliography - PRISM model checker, <https://www.prismmodelchecker.org/bib.php>, -. [Accessed: July 10, 2024].
- [13] K. A. Hoque, O. A. Mohamed, Y. Savaria, Towards an accurate reliability, availability and maintainability analysis approach for satellite systems based on probabilistic model checking, in: 2015 Design, Automation & Test in Europe Conference & Exhibition (DATE), 2015, pp. 1635–1640. doi:[10.7873/DATE.2015.0817](https://doi.org/10.7873/DATE.2015.0817).
- [14] Y. Lu, Z. Peng, A. Miller, T. Zhao, C. W. Johnson, How reliable is satellite navigation for aviation? checking availability properties with probabilistic verification, *Reliab. Eng. Syst. Saf.* 144 (2015) 95–116. URL: <https://doi.org/10.1016/j.ress.2015.07.020>. doi:[10.1016/J.RESS.2015.07.020](https://doi.org/10.1016/J.RESS.2015.07.020).
- [15] Z. Peng, Y. Lu, A. Miller, C. W. Johnson, T. Zhao, A probabilistic model checking approach to analysing reliability, availability, and maintainability of a single satellite system, in: Seventh UKSim/AMSS European Modelling Symposium, EMS 2013, 20–22 November, 2013, Manchester UK, IEEE, 2013, pp. 611–616. URL: <https://doi.org/10.1109/EMS.2013.102>. doi:[10.1109/EMS.2013.102](https://doi.org/10.1109/EMS.2013.102).
- [16] A. Baouya, B. Hamid, O. A. Mohamed, S. Bensalem, Model-based reliability, availability, and maintainability analysis for satellite systems with collaborative maneuvers via stochastic games, in: 2024 50th Euromicro Conference on Software Engineering and Advanced Applications (SEAA), 2024, pp. 27–34. doi:[10.1109/SEAA64295.2024.00014](https://doi.org/10.1109/SEAA64295.2024.00014).
- [17] M. Kwiatkowska, G. Norman, D. Parker, Stochastic Model Checking, Springer Berlin Heidelberg, Berlin, Heidelberg, 2007, pp. 220–270. URL: https://doi.org/10.1007/978-3-540-72522-0_6. doi:[10.1007/978-3-540-72522-0_6](https://doi.org/10.1007/978-3-540-72522-0_6).
- [18] A. D. Swain, H. E. Guttmann, Handbook of human-reliability analysis with emphasis on nuclear power plant applications. Final report, Technical Report, Sandia National Lab. (SNL-NM), Albuquerque, NM (United States), 1983. URL: <https://www.osti.gov/biblio/5752058>. doi:[10.2172/5752058](https://doi.org/10.2172/5752058).

¹HERMES-Design: <https://hermes-design.github.io/>

- [19] D. Nunes, J. Silva, F. Boavida, A Practical Introduction to Human-in-the-Loop Cyber-Physical Systems, IEEE Press, Wiley, 2018. URL: <https://books.google.fr/books?id=4qVFDwAAQBAJ>.
- [20] H. M. Prévost, et al., Neurological manifestations of hypothermia, Journal of Neurology, 2018. Analyse de l'impact de l'hypothermie sur le fonctionnement cérébral, provoquant des troubles cognitifs et psychologiques, notamment par la diminution de l'activité neuronale et la perturbation de la neurotransmission.
- [21] A. M. Ghadimi, J. L. Sapra, Hypothermia, 2023. URL: <https://www.ncbi.nlm.nih.gov/books/NBK545239/>, updated: August 28, 2023. In: StatPearls [Internet]. NCBI Bookshelf.
- [22] S. Kent, Sherman kent and the profession of intelligence analysis- declassified and approved document, <https://www.cia.gov/readingroom/docs/CIA-RDP93T01132R000100020036-3.pdf>, 2002. P. 55.
- [23] M. Kwiatkowska, G. Norman, D. Parker, G. Santos, Equilibria-based probabilistic model checking for concurrent stochastic games, in: Formal Methods – The Next 30 Years, Springer International Publishing, Cham, 2019, pp. 298–315.
- [24] P. Bhandari, C. Boyle, J. Gong, K. M. Y. Law, D. Creighton, Ongoing transformation of critical infrastructure systems as cyberphysical-human systems, in: 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2021, pp. 3342–3347. doi:10.1109/SMC52423.2021.9658996.
- [25] M. Dong, R. Ranjan, A. Y. Zomaya, M. Lin, Guest editorial on advances in tools and techniques for enabling cyber-physical-social systems—part i, IEEE Transactions on Computational Social Systems 2 (2015) 38–40. doi:10.1109/TCSS.2016.2527158.
- [26] N. Wiener, Cybernetics: Or Control and Communication in the Animal and the Machine, first edition ed., MIT Press, Cambridge, MA, 1948.
- [27] B. Kirwan, A Guide To Practical Human Reliability Assessment, 1st ed., CRC Press / Taylor & Francis, Boca Raton, FL, USA / London, UK, 1994.
- [28] D. Gertman, H. Blackman, J. Marble, C. Smith, R. Boring, The spar-h human reliability analysis method, 2004.
- [29] W. Si, L. Niu, Enhancing human reliability prediction in smart tower crane interfaces: A refined approach using simplified plant analysis risk–human reliability assessment and the decision making trial and evaluation laboratory–analytic network process, Buildings 14 (2024). URL: <https://www.mdpi.com/2075-5309/14/4/1083>. doi:10.3390/buildings14041083.
- [30] R. Alegre-Godoy, I. Stojkovic, Improving the availability of the sar/galileo return link service via network coding, in: 2014 7th ESA Workshop on Satellite Navigation Technologies and European Workshop on GNSS Signals and Signal Processing (NAVITEC), 2014, pp. 1–6. doi:10.1109/NAVITEC.2014.7045149.
- [31] I. Joo, S. Lee, J.-H. Kim, Study on development strategy for sar/galileo ground system in korea, in: 2007 International Conference on Control, Automation and Systems, 2007, pp. 2546–2549. doi:10.1109/ICCAS.2007.4406794.
- [32] A. Lewandowski, B. Niehoefer, C. Wietfeld, Performance evaluation of satellite-based search and rescue services: Galileo vs. cospas-sarsat, in: 2008 IEEE 68th Vehicular Technology Conference, 2008, pp. 1–5. doi:10.1109/VETECF.2008.69.
- [33] C.-O. Andrei, J. Johansson, H. Koivula, M. Poutanen, Signal performance analysis of the latest quartet of galileo satellites during the first operational year, in: 2020 International Conference on Localization and GNSS (ICL-GNSS), 2020, pp. 1–6. doi:10.1109/ICL-GNSS49876.2020.9115548.
- [34] PRISM Development Team (eds.), PRISM Manual, <https://www.prismmodelchecker.org/manual/ConfiguringPRISM/ComputationEngines>, Accessed: July 10, 2025.
- [35] PRISM Model Checker, PRISM - Publications, 2025. URL: <https://www.prismmodelchecker.org/casestudies/index.php#reliability>.
- [36] M. Kwiatkowska, G. Norman, D. Parker, Approximate symbolic model checking of continuous-time markov chains, in: Proceedings of the 14th International Conference on Concurrency Theory, Springer, 2002, pp. 51–65.
- [37] T. Chen, V. Forejt, M. Kwiatkowska, D. Parker, A. Simaitis, Automatic verification of competitive stochastic systems, in: Tools and Algorithms for the Construction and Analysis of Systems, volume 7214, Springer Berlin Heidelberg, 2012, pp. 315–330.
- [38] R. Alur, T. A. Henzinger, O. Kupferman, Alternating-time temporal logic, J. ACM 49 (2002) 672–713.
- [39] H. Hansson, B. Jonsson, A logic for reasoning about time and reliability 6 (1994) 512–535. URL: <https://dl.acm.org/doi/10.1007/BF01211866>. doi:10.1007/BF01211866.
- [40] S. Lichtenstein, J. R. Newman, Empirical scaling of common verbal phrases associated with numerical probabilities, Psychonomic Science 9 (1967) 563–564. URL: <https://doi.org/10.3758/BF03327890>. doi:10.3758/BF03327890.
- [41] M. Manning, M. Petit, A. Allal, M. C. Bruno, C. B. Field, S. J. N. O'Brien, H. J. Schellnhuber, G. Yohe, Guidance Note for Lead Authors of the IPCC Fourth Assessment Report on Addressing Uncertainties, 2007. URL: <https://www.ipcc.ch/site/assets/uploads/2018/02/ar4-uncertaintyguidancenote-1.pdf>, approved by the IPCC Panel in February 2007.
- [42] Cleveland Clinic, Hypothermia (Low Body Temperature), -. URL: <https://my.clevelandclinic.org/health/diseases/21164-hypothermia-low-body-temperature>, accessed on: , July 11, 2025.
- [43] J. Benezet-Mazuecos, B. Ibanez, J. Farre, Severe hypothermia showing osborn waves associated with transient atrial fibrillation and st segment depression, Heart 92 (2006) 1666. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC1861236/>.
- [44] J. M. Stocks, N. A. S. Taylor, M. J. Tipton, J. E. Greenleaf, Human physiological responses to cold exposure, Aviation, Space, and Environmental Medicine 75 (2004) 444–457. URL: <https://pubmed.ncbi.nlm.nih.gov/15152898/>. doi:10.3357/ASEM.1336.2004.
- [45] T. Deslarzes, V. Rousson, B. Yersin, B. Durrer, M. Pasquier, An evaluation of the swiss staging model for hypothermia using case reports from the literature, Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine 24 (2016) 16. URL: <https://doi.org/10.1186/s13049-016-0210-y>. doi:10.1186/s13049-016-0210-y.
- [46] E. A. Nye, J. R. Edler, L. E. Eberman, K. E. Games, Optimizing cold-water immersion for exercise-induced hyperthermia: An evidence-based paper, Medicine & Science in Sports & Exercise 47 (2015) 2464–2472. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5076277/>.

- [47] X. Xu, J. P. Meyer, W. R. Santee, Prediction of core temperature responses from multi-location skin temperatures in hypothermia and hyperthermia scenarios, *Extreme Physiology & Medicine* 3 (2014) A132. URL: <https://extremephysiolmed.biomedcentral.com/articles/10.1186/2046-7648-4-S1-A132>. doi:10.1186/2046-7648-4-S1-A132.
- [48] E. G. A. (GSA), Search and rescue (sar) - galileo service, <https://www.gsc-europa.eu/galileo/services/search-and-rescue-sar-galileo-service>, Accessed in February 2025.
- [49] E. U. A. for the Space Programme (EUSPA), Galileo search and rescue service definition document (sdd), <https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo-SAR-SDD.pdf>, Accessed in January 2025.
- [50] European Space Agency (ESA), Galileo General Introduction, -. URL: https://gssc.esa.int/navipedia/index.php?title=Galileo_General_Introduction, galileo Support Services Centre (GSSC).
- [51] M. Kwiatkowska, G. Norman, D. Parker, G. Santos, Prism-games 3.0: Stochastic game verification with concurrency, equilibria and time, in: *Computer Aided Verification*, Springer International Publishing, Cham, 2020, pp. 475–487.
- [52] Paper Artefacts Sources, <https://hermes-design.github.io/iot2025.html>, 2025.
- [53] A. Baouya, B. Hamid, L. Gürgen, S. Bensalem, Rigorous security analysis of rabbitmq broker with concurrent stochastic games, *Internet of Things* 26 (2024) 101161. URL: <https://www.sciencedirect.com/science/article/pii/S2542660524001021>. doi:<https://doi.org/10.1016/j.iot.2024.101161>.